

Kraftworks Career Hub - Backend Development Proposal

KRAFTWORKS

Backend Development Proposal

Career Hub Platform

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Prepared by: Backend Developer

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1. Executive Summary

This proposal outlines the complete backend development for Kraftworks Career Hub, a career platform for skilled trades professionals. The solution leverages modern serverless architecture to deliver a **scalable, cost-effective, and production-ready** system.

1.1 Project Overview

Kraftworks Career Hub requires a robust backend infrastructure to support AI-powered career development tools for trade professionals including HVAC technicians, electricians, welders, and other skilled workers.

1.2 Key Deliverables

- **Resume Review System** — AI-powered resume analysis with admin approval workflow
- **Interview Preparation** — Automated interview question generation based on job descriptions
- **Authentication System** — Enterprise-grade user authentication and authorization
- **Email Notifications** — Transactional email system for user communications
- **Admin Dashboard API** — Complete administrative control interface

1.3 Investment Summary

Item	Amount
Total Development Hours	20 hours
Hourly Rate	\$30 USD (Rp 495,000)
Total Project Cost	Rp 9,900,000
Monthly Infrastructure (est.)	Rp 8,250 – 90,750

2. Scope of Work

2.1 In-Scope Features

The following features are included in this development engagement:

Feature	Priority	Description
Resume Review	High	AI-powered feedback with approval workflow
Interview Prep	High	Question generation from job descriptions
Authentication	High	Clerk integration with role-based access
Email System	High	Transactional notifications via Resend
Admin Workflow	High	Approval queue and user management

2.2 Out-of-Scope (Deferred)

- Digital Career Fair functionality
- Blog and content management system
- Mobile application backend
- Payment processing integration

2.3 Technical Constraints

- Maximum 3 AI actions per user during beta period
- Single resume per user (PDF/DOCX, max 5MB)
- Admin approval required before feedback delivery

3. Technical Architecture

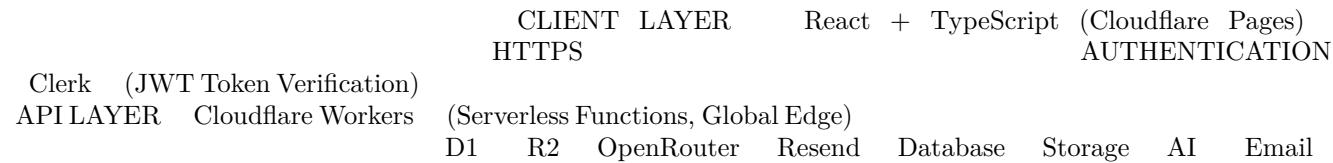
3.1 Technology Stack

The selected technology stack prioritizes **cost efficiency**, **global performance**, and **operational simplicity**.

Component	Technology	Justification
Authentication	Clerk	50K MAU free, pre-built UI
Serverless Functions	Cloudflare Workers	100K req/day free, edge-native
Database	Cloudflare D1	5GB free, SQLite compatible
File Storage	Cloudflare R2	Zero egress fees, S3-compatible

Component	Technology	Justification
AI Processing	OpenRouter	Multi-model, \$0.01/1M tokens
Email Service	Resend	3K emails/month free

3.2 System Architecture



3.3 Data Model

Core Entities

Entity	Purpose
<code>users</code>	User accounts synced from Clerk
<code>profiles</code>	Extended user information (trade, location)
<code>resumes</code>	Uploaded resume files and extracted text
<code>resume_feedback</code>	AI-generated feedback with approval status
<code>interview_sessions</code>	Interview prep sessions with job descriptions
<code>interview_questions</code>	Generated questions (technical, behavioral, situational)
<code>ai_usage</code>	Usage tracking for beta limits
<code>admin_users</code>	Admin role assignments
<code>email_logs</code>	Email delivery tracking

4. Feature Specifications

4.1 Resume Review System

User Flow

1. User uploads resume (PDF/DOCX, max 5MB)
2. System extracts text using native parser or OCR fallback
3. AI analyzes resume and generates structured feedback
4. Feedback stored with `pending_review` status
5. Admin notified via email for review
6. Admin approves/rejects feedback with optional notes
7. User notified when feedback is ready

Feedback Structure

Field	Description
<code>overall_score</code>	Numeric score 1-10
<code>strengths</code>	3-4 key resume strengths
<code>improvements</code>	3-4 areas needing improvement
<code>suggestions</code>	2-3 specific recommendations
<code>trade_suggestions</code>	Trade-specific advice (HVAC, electrical, etc.)
<code>actionable_steps</code>	3-4 concrete improvement actions

4.2 Interview Preparation System

User Flow

1. User enters job description (min 10 characters)
2. System retrieves user's resume for context
3. AI generates tailored interview questions
4. Questions categorized by type and stored
5. Results displayed immediately to user

Question Categories

- **Technical** — Trade-specific technical knowledge questions
- **Behavioral** — Past experience and behavior-based questions
- **Situational** — Hypothetical scenario-based questions

4.3 Authentication & Authorization

Role-Based Access Control

Role	Permissions
anonymous	Waitlist signup, public pages
user	Resume upload, AI feedback (3x limit), interview prep
reviewer	View pending feedback, approve/reject, add notes
admin	All reviewer permissions + user management, analytics
super_admin	All admin permissions + system configuration

4.4 Email Notification System

Email Type	Recipient	Trigger
Welcome	User	Account creation
Feedback Pending	Admin	AI generates feedback
Feedback Approved	User	Admin approves feedback
Feedback Rejected	User	Admin rejects feedback
Interview Ready	User	Questions generated
Waitlist Confirmation	User	Waitlist signup

5. API Specification

5.1 Base URL

<https://api.kraftworks.careers/api/v1>

5.2 Endpoint Summary

Method	Endpoint	Description
User Endpoints		
GET	/users/me	Get current user profile
PATCH	/users/me	Update user profile
Resume Endpoints		
GET	/resumes	Get user's resume
POST	/resumes	Upload resume
DELETE	/resumes/:id	Delete resume

Method	Endpoint	Description
POST	/resumes/:id/feedback	Generate AI feedback
Interview Endpoints		
GET	/interviews	List interview sessions
POST	/interviews	Generate questions
GET	/interviews/:id	Get specific session
Admin Endpoints		
GET	/admin/feedback	List pending feedback
PATCH	/admin/feedback/:id	Approve/Reject feedback
GET	/admin/users	List all users
GET	/admin/analytics	Usage analytics

6. Cost Analysis

6.1 Free Tier Allocation

Service	Free Tier	Notes
Clerk Authentication	50,000 MAU	Full feature access
Cloudflare Workers	100K requests/day	Global edge deployment
Cloudflare D1	5M reads/day, 5GB	SQLite database
Cloudflare R2	10GB, 1M operations	Zero egress fees
Resend Email	3,000 emails/month	Full API access

6.2 AI Processing Costs

Action	Est. Tokens	Cost
Resume Feedback Generation	2,000	\$0.0001
Interview Question Generation	3,000	\$0.00015
Total per User (3 actions)	15,000	\$0.00075

6.3 Monthly Cost Projections

Users	Workers	D1	R2	AI	Email	Total
100	\$0	\$0	\$0	\$0.50	\$0	\$0.50
500	\$0	\$0	\$0	\$2.50	\$0	\$2.50
1,000	\$0	\$0	\$0.50	\$5.00	\$0	\$5.50
5,000	\$5	\$2	\$3	\$25	\$20	\$55
10,000	\$10	\$5	\$8	\$50	\$20	\$93

Key Insight: The serverless architecture enables near-zero infrastructure costs for up to 1,000 users, with linear scaling thereafter. Monthly costs remain under \$100 even at 10,000 users.

7. Project Timeline

7.1 5-Day Development Sprint

Day	Hours	Deliverables
Day 1	4 hours	Foundation & Setup • Cloudflare project initialization (Workers, D1, KV) • Clerk authentication configuration • Database schema creation • API routing structure
Day 2	4 hours	Resume Review Core • File upload to R2 storage • PDF/DOCX text extraction • OpenRouter AI integration • Feedback generation endpoint
Day 3	4 hours	Interview Prep & Admin • Interview question generation • Admin approval workflow • Email notifications setup • Usage tracking implementation
Day 4	4 hours	Integration & Testing • Frontend-backend integration • Error handling & edge cases • Rate limiting implementation • Security review
Day 5	4 hours	Deployment & Polish • Production deployment • DNS configuration • Monitoring setup • Documentation handoff

7.2 Milestone Schedule

Milestone	Date	Status
Project Kickoff	March 26, 2026	Pending
Foundation Complete	March 26, 2026	Pending
Core Features Complete	March 28, 2026	Pending
Testing Complete	March 30, 2026	Pending
Final Delivery	April 1, 2026	Pending

8. Investment & Terms

8.1 Project Quotation

Description	Hours	Amount (IDR)
Day 1: Foundation & Setup	4	Rp 1,980,000
Day 2: Resume Review Core	4	Rp 1,980,000
Day 3: Interview Prep + Admin	4	Rp 1,980,000
Day 4: Integration + Testing	4	Rp 1,980,000
Day 5: Deployment + Polish	4	Rp 1,980,000
TOTAL	20	Rp 9,900,000

Rate: \$30 USD/hour = Rp 495,000/hour

Exchange Rate: 1 USD = Rp 16,500 (approximate)

8.2 Payment Terms

Milestone	%	Amount	Due Date
Project Start	50%	Rp 4,950,000	March 26, 2026
Project Completion	50%	Rp 4,950,000	April 1, 2026
Total	100%	Rp 9,900,000	—

8.3 Deliverables Included

- Backend infrastructure (Workers, D1, R2)
- Clerk authentication integration
- Resume review with AI feedback
- Interview prep question generation
- Admin approval workflow
- Email notification system
- Database schema & migrations
- API documentation
- Production deployment
- 1 week post-delivery support

9. Terms & Conditions

9.1 Scope Management

- Changes to scope require written approval and may affect timeline and pricing
- Out-of-scope items can be added as separate engagements
- Bug fixes related to delivered features are covered for 7 days post-delivery

9.2 Intellectual Property

- All code developed becomes client property upon final payment
- Third-party libraries retain their respective licenses
- Documentation and technical specifications become client property

9.3 Confidentiality

- All project information is treated as confidential
- No project details shared with third parties without written consent
- Source code and credentials handled with industry-standard security practices

9.4 Acceptance Criteria

The project will be considered complete when:

1. All in-scope features are functional and deployed
2. API endpoints respond correctly per specification
3. Authentication and authorization work as designed
4. Email notifications are delivered successfully
5. Admin approval workflow operates correctly
6. Documentation is provided for all endpoints

10. Appendix

10.1 Technology References

- **Clerk Documentation:** <https://clerk.com/docs>

- **Cloudflare Workers:** <https://developers.cloudflare.com/workers>
- **Cloudflare D1:** <https://developers.cloudflare.com/d1>
- **Cloudflare R2:** <https://developers.cloudflare.com/r2>
- **OpenRouter API:** <https://openrouter.ai/docs>
- **Resend:** <https://resend.com/docs>

10.2 Contact Information

Backend Developer

Email: [developer@email.com]

Response time: Within 24 hours

10.3 Approval

Client Approval Name:

Date:

Signature:

Developer Acknowledgment Name:

Date:

Signature:

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